

Juan Enrique Ayala Zapata

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EDUCATION

Monterrey's Institute of Technology and Higher Education (Tecnológico de Monterrey) <i>Bachelor of Science in Computer Science, Minor in Cybersecurity</i> • GPA: 96.35/100 (3.9/4.0) Academic Talent Scholarship Recipient • Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, IoT Implementation, Computational Thinking	Querétaro, MX <i>Expected: June 2027</i>
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EXPERIENCE

CTO & Co-Founder <i>Neural Harvest – AgTech Startup</i> • Selected for Dedalus Break In Program residency in San Francisco, joining 20 founders from 13 countries in month-long AI startup accelerator • Leading development of AI-powered precision agriculture platform combining satellite imagery analysis with proprietary ML crop simulation engines • Architecting scalable microservices infrastructure processing multi-spectral satellite data with hybrid database systems and containerized deployment • Managing cross-functional team implementing enterprise-grade RBAC, real-time IoT monitoring, and automated field communication systems	February 2025 – Present Querétaro, MX
Back-End Developer <i>ceams.co – Digital Agency</i> • Developed scalable back-end infrastructure using Node.js and MySQL , optimizing data collection and storage systems for client web applications • Improved website performance by 40% through server-side optimizations, efficient database query implementation, and caching strategies • Enhanced SEO performance and search engine visibility through technical optimizations, schema markup, and page speed improvements	March 2024 – May 2025 Querétaro, MX

PROJECTS

START Hack - OrbitGrow Martian Greenhouse <i>Python, FastAPI, React, AWS SAM/Lambda</i> • Architected autonomous multi-agent AI system with 5 specialized agents (Nutrition, Environment, Crisis, Planner, Vision) orchestrated via Sol-loop coordinator to manage a Martian greenhouse sustaining 4 astronauts over a 450-day mission • Built real-time simulation engine modeling crop growth, weather drift, and crisis events across 12 greenhouse plots, with all agent decisions grounded in the Syngenta MCP Knowledge Base via streamable HTTP endpoint • Engineered full-stack platform with FastAPI backend, React digital twin dashboard, and WebSocket live updates, deployed on AWS using SAM with Lambda functions, API Gateway, and 6 DynamoDB tables	March 2026
TreeHacks - Sentinel AI Security Scanner <i>Python, Next.js, Gemini, Playwright, Supabase</i> • Architected autonomous penetration testing platform deploying swarm of 10 specialized AI agents for concurrent web security scanning with coordinated reconnaissance, exploitation, and analysis phases • Engineered Flask-based orchestration engine with asyncio managing concurrent agent execution across attack surface mapping, SQLi/XSS fuzzing, CORS/header auditing, and autonomous Playwright -powered Red Team agent • Built real-time Next.js 16 dashboard with Supabase Realtime event streaming, severity-weighted security scoring (A-F), and Gemini -generated remediation reports with OWASP references	February 2026
Prometheus - AI-Powered Recruitment Platform <i>Python, Flask, Next.js, Google Gemini 2.5, Firebase</i> • Architected conversational AI recruitment system using Gemini 2.5 Flash with ADK, implementing progressive filtering across multi-turn dialogues for semantic candidate matching beyond keyword search • Engineered full-stack platform with Flask REST API , Next.js 15 , and Firebase Firestore , integrating Twilio WhatsApp API for 98% open rates vs 20% email, with CV parsing and automated job description generation • Developed semantic matching with transferable skills analysis (Vue.js → React equivalence), 0-100 compatibility scoring, reducing candidate discovery from 3-5 days to 2 minutes	October 2025
Bloombly - Wildflower Bloom Prediction Platform <i>Python, Flask, TensorFlow, Google Earth Engine</i> • Engineered ML platform predicting wildflower blooms using NASA satellite data (MODIS, SMAP) with 31+ ecological features including spring phenology detection, growing degree days, and soil water stress analysis • Developed Gradient Boosting classifier achieving 0.72-0.85 ROC-AUC with feature engineering combining spatial, temporal, and environmental variables across 73 years of climate data, including cherry blossom model with ±3 days accuracy • Built full-stack app with Flask REST API , Google Earth Engine integration, and interactive 3D globe visualization supporting global bloom predictions with 90-day forecasting	October 2025

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL/MySQL, HTML/CSS, R, MATLAB Frameworks & Libraries: FastAPI, Flask, Next.js, React, TensorFlow, Scikit-learn, Pandas, NumPy, Node.js, Express.js Tools & Technologies: AWS (SAM, Lambda, DynamoDB), Playwright, Supabase, Docker, GCP, Google Earth Engine, Git, PostgreSQL, Firebase Specializations: AI Agent Systems, Penetration Testing, Cybersecurity, Machine Learning, Full-Stack Development, Real-time Systems, Web Security
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